

1. Groups

Cyclic. $C_n = \langle g \mid g^n = 1 \rangle$. G is cyclic iff $G = \langle g \rangle$ for some $g \in G$.

Dihedral. $D_n = \langle r, s \mid r^n = s^2 = 1, srs = r^{-1} \rangle$, $|D_n| = 2n$.

Direct Product. For $(g, h) \in G \times H$: $o((g, h)) = \text{lcm}(o(g), o(h))$.

Lagrange. If G is finite and $H \leq G$, then $|H|$ divides $|G|$. Also, $o(g)$ divides $|G|$ for all $g \in G$.

Cauchy. If G is finite and prime $p \mid |G|$, then $\exists g \in G$ with $o(g) = p$.

Prime Order. If $|G| = p$ prime, then G is cyclic (hence abelian).

Normal Subgroup. $H \triangleleft G$ iff $gHg^{-1} = H$ for all $g \in G$.

Index 2. If $[G : H] = 2$, then $H \triangleleft G$ (index 2 subgroups are always normal).

Ex: $|G| = 10$. By Cauchy, $\exists$ elements of order 2 and 5. Each generates a cyclic subgroup. The subgroup of order 5 has index $10/5 = 2$, so it is normal.

2. Group Actions

Orbit. $\text{Orb}(x) = \{g \cdot x : g \in G\}$ (all elements reachable from x).

Stabilizer. $\text{Stab}(x) = \{g \in G : g \cdot x = x\}$ (elements fixing x).

Orbit-Stabilizer. If G finite, then $|G| = |\text{Orb}(x)| \cdot |\text{Stab}(x)|$.

Centralizer. $C_G(g) = \{h \in G : hg = gh\}$ (elements commuting with g).

3. Finite Abelian Groups

Structure Theorem. Every finite abelian group is isomorphic to a direct product of cyclic groups of prime power order:

$$G \cong \mathbb{Z}_{p_1^{e_1}} \times \mathbb{Z}_{p_2^{e_2}} \times \cdots \times \mathbb{Z}_{p_k^{e_k}}$$

Cyclic Test. $\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn}$ iff $\text{gcd}(m, n) = 1$.

Counting. $|G| = p_1^{a_1} \cdots p_r^{a_r}$ implies number of abelian groups $= \prod p(a_i)$ where $p(a_i)$ = partitions of a_i .

Elementary Divisors. The prime power factors $\{p_i^{e_{ij}}\}$.

Invariant Factors. $d_1 \mid d_2 \mid \cdots \mid d_r$ with $G \cong \mathbb{Z}_{d_1} \times \cdots \times \mathbb{Z}_{d_r}$.

Order Check. G contains element of order n iff $n \mid d_r$ (largest invariant factor).

4. Symmetric Groups

Definition. S_n = all bijections $\{1, \dots, n\} \rightarrow \{1, \dots, n\}$, with $|S_n| = n!$.

Cycle Notation. $(a_1 a_2 \cdots a_k)$: $a_i \mapsto a_{i+1}$, $a_k \mapsto a_1$.

Order. $o(\sigma) = \text{lcm}(\text{cycle lengths})$ when written in disjoint cycle form.

Cyclic Subgroup. $C_k \leq S_n$ exists iff k can be written as lcm of positive integers summing to $\leq n$.

5. Rings and Domains

FEPIC Chain. Field $\subset$ Euclidean Domain $\subset$ PID $\subset$ Integral Domain $\subset$ Comm. Ring.

Integral Domain. Commutative ring with $1 \neq 0$, no zero divisors: $ab = 0$ implies $a = 0$ or $b = 0$.

Zero Divisor. $a \neq 0$ with $\exists b \neq 0$: $ab = 0$.

PID. Every ideal is principal: $I = (a)$.

Euclidean Domain. Has division algorithm (e.g., $\mathbb{Z}$, $F[X]$).

Examples:

- $\mathbb{Z}$, $F[X]$: ED, PID
- $\mathbb{Z}/n\mathbb{Z}$: field iff n prime
- $\mathbb{Z}[X]$: domain, not PID
- $\mathbb{Z} \times \mathbb{Z}$: zero divisors

6. Fields

Definition. Commutative ring where every nonzero element has inverse.

Finite Field. $|\mathbb{F}| = p^n$ for prime p ; unique up to isomorphism.

Existence. $\mathbb{F}_q$ exists iff $q = p^n$. No $\mathbb{F}_{12}$, $\mathbb{F}_{18}$.

$\mathbb{Z}/n\mathbb{Z}$. Field iff n prime.

Characteristic. Smallest $n > 0$ such that $n \cdot 1 = 0$. For $\mathbb{F}_p$: $\text{char} = p$. For any ring A : $\text{char}(A[X]) = \text{char}(A)$.

7. Polynomials and Ideals

$\mathbb{F}_p[X]$. Polynomials over field $\mathbb{F}_p$. Reduce coefficients mod p ; remember $-1 \equiv p - 1$.

GCD. In a Euclidean domain, use Euclidean algorithm; last nonzero remainder (make monic).

Extended Euclidean. Find r, s such that $\text{gcd}(f, g) = f \cdot r + g \cdot s$.

Ideal Operations. In a PID:

- $(a)(b) = (ab)$
- $(a) \cap (b) = (\text{lcm}(a, b))$
- $(a) + (b) = (\text{gcd}(a, b))$

Containment. $(p) \subseteq (a)$ iff $a \mid p$.

8. Quotient Rings

Definition. For ring R and ideal $I \subseteq R$: R/I with $[a] + [b] = [a + b]$, $[a][b] = [ab]$.

Computing. In $F[X]/(p(X))$ where F is a field: set $p(X) = 0$, reduce all polynomials mod $p(X)$.

Field Criterion. For field F : $F[X]/(p(X))$ is a field iff $p(X)$ is irreducible over F .

9. Irreducibility

Degree ≤ 3 . For $f \in F[X]$ with $\deg f \leq 3$: f irreducible over F iff f has no roots in F .

Rational Root. For $f = a_n X^n + \cdots + a_0 \in \mathbb{Z}[X]$: if $\frac{r}{s}$ (reduced) is a root, then $r \mid a_0$ and $s \mid a_n$.

Eisenstein. For $f = a_n X^n + \cdots + a_0 \in \mathbb{Z}[X]$: if $\exists$ prime p with $p \nmid a_n$, $p \mid a_i$ for all $i < n$, $p^2 \nmid a_0$, then f is irreducible over $\mathbb{Q}$.

Reduction mod p . For $f \in \mathbb{Z}[X]$: if $\bar{f} \in \mathbb{F}_p[X]$ is irreducible with $\deg \bar{f} = \deg f$, then f is irreducible over $\mathbb{Q}$.

10. Euler's Totient

Definition. $\varphi(n)$ = count of integers $1 \leq a \leq n$ with $\text{gcd}(a, n) = 1$.

Formulas:

- Prime power: $\varphi(p^k) = p^{k-1}(p - 1) = p^k - p^{k-1}$
- Multiplicative: $\varphi(mn) = \varphi(m)\varphi(n)$ when $\text{gcd}(m, n) = 1$
- General: $\varphi(n) = n \cdot \prod_{p \mid n} \left(1 - \frac{1}{p}\right)$ (product over distinct primes)

Units. $(\mathbb{Z}/n\mathbb{Z})^\times = \{[a] : \text{gcd}(a, n) = 1\}$, has order $\varphi(n)$.

Euler's Theorem. If $\text{gcd}(a, n) = 1$, then $a^{\varphi(n)} \equiv 1 \pmod{n}$.

Fermat's Little. If p prime and $\text{gcd}(a, p) = 1$, then $a^{p-1} \equiv 1 \pmod{p}$. Also $a^p \equiv a \pmod{p}$ for any a .

Inverse. If $\text{gcd}(a, n) = 1$, use Extended Euclidean: $ax + ny = 1$ implies $a^{-1} \equiv x \pmod{n}$.

11. Chinese Remainder Theorem

Isomorphism. $\text{gcd}(m, n) = 1$ implies $\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$.

System. For pairwise coprime $n_1, \dots, n_k$: system $x \equiv a_i \pmod{n_i}$ has unique solution mod $N = \prod n_i$.

Formula. $x = \sum a_i M_i y_i$ where $M_i = N/n_i$, $y_i = M_i^{-1} \pmod{n_i}$.

Non-Coprime. Solvable iff $a_i \equiv a_j \pmod{\text{gcd}(n_i, n_j)}$ for all i, j .

12. Tips and Common Mistakes

Normality Check. To verify $H \triangleleft G$, only check $gHg^{-1} = H$ for generators of G .

$\mathbb{Z}[X]$ is NOT a PID: $(2, X)$ is not principal.

$\mathbb{Z}/8\mathbb{Z}$ is NOT a field: 8 is not prime.